Robust Flash Deblurring

Shaojie Zhuo Dong Guo Terence Sim
School of Computing, National University of Singapore, 117417
{zhuoshao,guodong,tsim}@comp.nus.edu.sg

Abstract

Motion blur due to camera shake is an annoying yet common problem in low-light photography. In this paper, we propose a novel method to recover a sharp image from a pair of motion blurred and flash images, consecutively captured using a hand-held camera. We first introduce a robust flash gradient constraint by exploiting the correlation between a sharp image and its corresponding flash image. Then we formulate our flash deblurring as solving a maximum-a-posteriori problem under the flash gradient constraint. We solve the problem by performing kernel estimation and non-blind deconvolution iteratively, leading to an accurate blur kernel and a reconstructed image with fine image details. Experiments on both synthetic and real images show the superiority of our method compared with existing methods.

1. Introduction

When taking photographs under low-light conditions using a hand-held camera, the captured images usually suffer from motion blur caused by camera shake within a long exposure time. Such a blurred image is usually modeled as the convolution of a sharp image with a motion blur kernel, which describes the camera motion path during exposure. The deblurring problem is a deconvolution process to recover a visually plausible sharp image from the blurred image.

Single image deblurring is a challenging problem, where both the blur kernel and the sharp image are unknown. Recently, great progress [6, 17] has been made by enforcing strong priors on both sharp images and motion blur kernels. Good results can be obtained when the kernel is small. However, single image deblurring methods are sensitive to noise and suffer from deconvolution artifacts such as ringing effects. Some methods seek to utilize the correlation among multiple images of the same scene for image deblurring. These methods use a blurred and noisy image pair [21] or two blurred images [3] as their inputs. Such image pairs can compensate each other and yield accurate kernel estimation and deconvolution results. However, some of the high-frequency details are lost in the blur process, which cannot be recovered by these methods.

(a) Blurred image (b) Flash image (c) Deblurring result

Figure 1. Flash deblurring using a pair of blurred and flash images.

In this paper, we propose a novel method to recover a high-quality sharp image by combining a blurred image and a corresponding flash image. Fig. 1 shows an example of a blurred and flash image pair. The blurred image records the ambient light, but it's details are blurred. The flash image is captured using a short exposure time. It is sharp and contains high-frequency details of the scene.

We investigate the correlation between a sharp image and its corresponding flash image and propose the flash gradient constraint. By integrating the flash gradient constraint into a maximum-a-posteriori (MAP) framework, we solve the optimization problem in an iterative way which alternates between kernel estimation and sharp image reconstruction. We demonstrated that our kernel estimation is accurate and robust to noise. Our sharp image reconstruction is able to greatly reduce noise and ringing artifacts without lost of fine image details, leading to a high quality sharp ambient image. The ambient illumination is preserved and the flash artifacts are properly handled. Fig. 1(c) shows one example of our flash deblurring results.

The key contribution of this paper is the application of flash photography to image deblurring. By using the flash image, our method can significantly reduce deblurring artifacts while keeping fine image details, producing high quality deblurring results. Moreover, our input images can be

978-1-4244-6983-3/10/$26.00 ©2010 IEEE

taken using only a conventional, hand-held camera with flash and thus our method is very practical for low-light photography.

2. Related Work

Image deblurring is well studied and a comprehensive survey can be found in [11]. Image deblurring can be separated into two steps: kernel estimation and sharp image reconstruction. Early methods for kernel estimation assume the motion to be a simple parametric form. Recently, great progress has been made to handle more general motions. Fergus *et al*. [6] used a variational Bayes inference method with natural image statistics to estimate the motion blur kernel. Jia *et al*. [8] investigated the relationship between object boundary transparency and the image motion blur and estimated the blur kernel from the alpha matte of motion objects. Shan *et al*. [17] formulated the deblurring problem as an MAP problem and proposed a high-order derivatives image noise model and a local image prior to avoid trivial solution. These methods are able to obtain an accurate kernel when the blur is small. However, when the blur kernel is large and noise present in the image, kernel estimation becomes inaccurate and unreliable. In contrast, our method can handle large motion blur due to camera shake and is robust to noise.

Given the estimated blur kernel, the second step is to reconstruct a sharp image from the blurred image. This step is also known as the non-blind image deconvolution. The most common methods are the Wiener filter and the Richardson-Lucy (RL) deconvolution [14]. However, these methods suffer from deconvolution artifacts such as amplified noise and ringing artifacts. In order to reduce artifacts, regularization methods have been proposed, such as total variation (TV) regularization [20] and Bilateral regularization [22]. Recent research in natural image statistics shows that natural image gradients follow a heavy-tailed distribution. The sparse gradient constraint [12] derived from this distribution encourages piecewise smoothness in the image while keeping salient structures of the image. In wavelet domain, similar regularization has been performed on the wavelet coefficients [15]. These regularization methods share a common weakness, *i.e*. the lost of fine image details is inevitable since they are concealed in the noise and ringing artifacts and cannot be separated properly. Readers can refer to [13] for a quantitatively evaluation of several blind deconvolution methods.

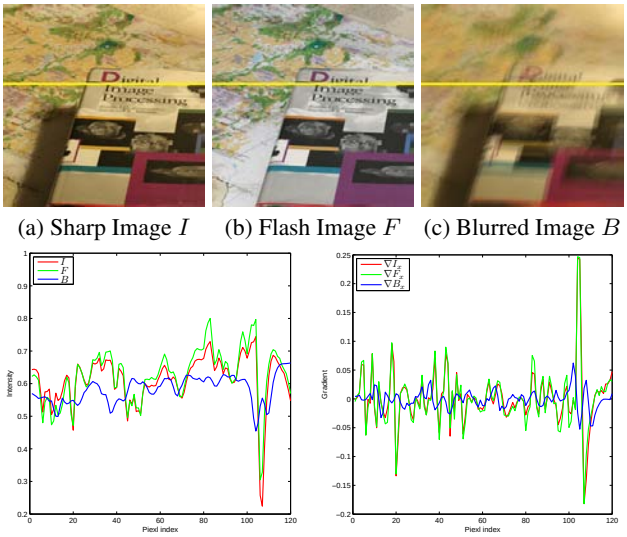
Methods using multiple images and additional hardware have also been proposed. Additional images or hardware provide more information on image details and camera motion, which helps to better estimate a blur kernel and reduce deconvolution artifacts. These methods either used a pair of blurred and noisy images [21], or used two blurred images [3]. Color transfer [19] has also been applied to transfer the color from a blurred image to its corresponding noisy image without kernel estimation. In [2, 18], an low-resolution video camera is connected to a high-resolution camera to record video sequence during exposure, which later is used to estimate the blur kernel. Compared with single image deblurring, these methods outperformed both in kernel estimation and sharp image reconstruction. However, some image details are still lost due to non-invertible motion blur. Our method uses the flash image which contains high frequency details of the scene. The lost image details can be recovered from the flash image. Besides, our method does not require any additional hardware.

Our work also relates to the flash/no-flash technique. Agrawal *et al*. [1] used flash/no-flash images to remove flash artifacts by projecting the flash image gradient to the ambient image gradients. Using cross bilateral filter and detail transfer, Petschnigg *et al*. [16] used the flash image to denoise the ambient image, while Eisemann and Durand [5] transfer the ambient tone to the flash image. To avoid intrusive burst of light produced by visible flash, Krishnan and Fergus [10] invented the dark flash that uses infra-red and ultra-violet light. They exploited the correlation between nearby spectral bands to denoise the ambient image using the dark flash image. These methods produce impressive results but require good alignment between the two images, which limits their application in practice. By comparison, our method does not require exact alignment of images. A pair of blurred and flash images can be taken using a conventional hand-held camera with flash. Thus, our method is more practical for low-light photography.

3. Flash Gradient Constraint

Our input is a pair of images: a blurred image B and its corresponding flash image F. Both images are taken using the same aperture setting so that the defocus blur difference of the two images can be neglected. B is blurred but preserves the ambient illumination. F is sharp and contains high-frequency image details, but ambient illumination is destroyed by flash and it may contain some flash artifact such flash shadows and specularities. We want to use both images to reconstruct a sharp image I, which combines the ambient illumination from B and high-frequency image details from F. To achieve this, we first investigate the correlation between I and F. Agrawal *et al*. [1] observed that the gradient orientations of ambient and flash images are coherent and use it to remove artifacts in flash photography. Previous flash/no-flash denoising methods utilize the similarity of local image structures between flash/no-flash images. Krisnan and Fergus [10] found that the gradient difference between two near spectral images follows a sparse distribution. Based on their findings, we investigate the correlation of a sharp image and its corresponding flash image.

As I and F are in the same spectral band, a stronger cor-



(a) Sharp Image I (b) Flash Image F (c) Blurred Image B

(d) Intensity Plot (e) Gradient Plot

Figure 2. Flash gradient constraint. (d)(e) show 1D scanlines (100th row) of intensities and gradients in R channel of the three images. The intensity I, B and F are different to each other, while ∇I is close to ∇F.

relation can be found. To show the correlation, we took a sharp image I and its corresponding flash image F using a tripod. I is then blurred to generate the blurred image B. In Fig. 2(d)(e), we show the intensities and gradients of a 1D scanline (100th row) in the three images. As we can see from the figure, the intensities of I, B and F are quite different from each other. In the gradient domain, ∇B and ∇F are quite different, but ∇I and ∇F are aligned well and close to each other. It has been demonstrated [1] that small change in the image gradient does not affect the visual appearance of an image. Hence, to reduce noise and ringing artifacts, we proposed a constraint to make sure that the gradients of our reconstructed sharp image are close to the gradients of the flash image F, while allowing the magnitude to differ at some regions such as flash artifacts and ambient shadows. To reject those outliers as well as noise, we use the following cost function to measure the difference of their gradients,

$$E_f(I) = \rho(\nabla I - \nabla F), \tag{1}$$

where $\rho(x) = log\left(1 + \frac{1}{2}\left(\frac{x}{\epsilon}\right)^2\right)$ is the Lorentzian robust estimator [7] and ϵ is a predefined constant. Robust estimation has been applied and shows its effectiveness in many computer vision problems [7, 9].

The robust flash gradient constraint encourages the gradients of reconstructed image to be close to those in F, while at the locations of flash artifacts or ambient shadows, it allow their gradient to differ to avoid flash artifacts and keep ambient illumination. If we use ℓ_2 norm to measure the gradient differences, the gradients of reconstructed image should match ∇F closely, which may cause artifacts at regions of flash artifacts and strong flash edges.

4. Flash Deblurring

We formulate the flash deblurring problem into an MAP framework and separate it into two steps: kernel estimation and sharp image reconstruction.

4.1. Problem formulation

Given the blurred image B and flash image F, our goal is to estimate a blur kernel K and a sharp image I, so that I, K and B can be represented by the convolution model and the gradients of I are close to those in F. This problem can be formulated as solving the following optimization problem:

$$\begin{aligned} &\arg\max_{I,K} P(I,K|B,F) \\ &= \arg\max_{I,K} P(B|I,K)P(I|F)P(K) \\ &= \arg\min_{I,K} L(B|I,K) + L(I|F) + L(K), \end{aligned} \tag{2}$$

where $L(\cdot) = -log(P(\cdot))$. Here, we assume that B is conditionally independent of F given I and K, I and K are independent conditioned on F, and K is independent of F. These assumptions are commonly made by most image deblurring methods. Our cost function has three terms: the likelihood term, the flash constraint term defined in the previous section and the kernel prior term. We then define the other two terms respectively.

The convolution model for image deblurring is $B = I \otimes K + n$. When modeling the image noise n as a set of independent and identically distributed (i.i.d.) Gaussian noise, we get the following likelihood term

$$L(B|I,K) = \|I \otimes K - B\|^2. \tag{3}$$

A motion blur kernel describes the path of the camera shake during exposure, it is commonly observed to be sparse and its distribution can be fitted by a hyper-laplacian distribution. Therefore, we define the kernel prior term as follows.

$$L(K) = \lambda_k |K|^\alpha, \tag{4}$$

where $\alpha \leq 1$ and we use $\alpha = 0.8$ in this paper. Having defined the three terms in our cost function, we can now rewrite Eq. (2) as,

$$\arg\min_{I,K} \|I \otimes K - B\|^2 + \lambda_f \rho(\nabla I - \nabla F) + \lambda_k |K|^\alpha, \tag{5}$$

where λ_f and λ_k are used to balance the three terms. The value of λ_f typically range from 0.001 to 0.1, depending on the noise in the blurred image. λ_k is usually set to 0.01.

Without direct optimization on the cost function using the entire images I and F, our deblurring method can be

separated into two steps: we use a selected patch of I and F to estimate the blur kernel. Then, we perform non-blind deconvolution on the whole image to reconstruct a sharp image. There are three benefits for doing this. First, we can select a good patch to avoid obvious flash artifacts which may affect the kernel estimation. Second, if the size of the selected patch is much smaller than that of the original image, the computation time for kernel estimation is greatly reduced. Finally, in the non-blind deconvolution step, the flash artifacts can be detected and handled correctly.

4.2. Kernel estimation

To avoid confusion, we still use B and F to denote the selected patches of the blurred and flash images respectively. We then optimize Eq. (5) by iteratively estimating I and K. When K is fixed, we can estimate I by solving the optimization problem:

$$\arg\min_{I} \|I \otimes K - B\|^2 + \lambda_f \rho(\nabla I - \nabla F). \tag{6}$$

We solve this using iterative re-weighted least squares (IRLS) [12]. IRLS performs the optimization iteratively by solving a weighted least squared problem. At each iteration, the weight for each pixel is defined as :

$$w_i = \frac{2}{2\epsilon^2 + (\nabla I_i - \nabla F_i).^2}, \tag{7}$$

where i denotes the pixel location. After estimating I, we fix I and then estimate K by solving:

$$\arg\min_{K} \|I \otimes K - B\|^2 + \lambda_k |K|^\alpha. \tag{8}$$

This can also be solved using IRLS. The two steps are alternated until the change in K is small enough.

To avoid local minimum when the blur kernel is large, the kernel estimation is performed in a coarse-to-fine manner in the scale space. At the coarsest scale, the blur kernel is initialized as a 3×3 delta kernel. The converged values of K and I at each scale are upsampled as the initialization values for the next scale.

4.3. Sharp image reconstruction

With the estimated blur kernel, we want to reconstruct a sharp image from the blurred image with assistance from the flash image. In general, directly solving the optimization problem in Eq. (6) will get reasonably good results. This is because our robust gradient constraint can not only suppress deconvolution artifacts and provide fine image details, but also properly handle flash artifacts such as shadows and specularities. The gradients in flash artifact regions appear only in the flash image F, leading to large gradient differences $\nabla I - \nabla F$. Hence, during the optimization, small weights are give to those gradient differences,

Single Image Deconvolution Our Method

Figure 3. Over-saturated artifacts. The single image deblurring method [12] suffers from over-saturated regions in the blurred image. Our sharp image reconstruction method can handle this problem by automatically detecting the over-saturated regions.

which prevent ∇I from being close to ∇F in flash artifacts regions. However, some artifacts may arise if λ_f is set to be large to suppress large noise or ringing artifacts. Thus, we need to weight flash gradient constraint less in these regions. Besides, over-saturated regions contains no blur information and cause artifacts in the deblurred results, which is a common problem for previous deblurring methods [6, 17]. Fig. 3 shows an example of the artifact. Our method can remove this artifact by increasing the flash gradient constrains in these regions. To achieve this, we build a mask image M to change the weight of flash gradient constraint locally. The mask image M is defined as follows,

$$M = \begin{cases} 20, & \text{for over-saturated regions in } B \\ 0, & \text{for flash artifact regions} \\ 1, & \text{for other regions.} \end{cases} \tag{9}$$

We blur the mask image M to make the values change smoothly from one region to another. The over-saturated regions in both B and F can be detected by finding the regions in which the pixel intensities are larger than 95% of the maximum intensity. For flash shadows detection, we use estimated blur kernel and single image deconvolution method to get a sharp image and combine it with flash image to detect shadow regions, but we found that previous method [16] for artifact detection is not effective enough, especially for soft shadows detection. Hence, a simple markup can be provided to specify the shadows regions. In many situations, the shadow regions still contains useful gradients, so we only need to mark the shadow edges.

Now we can reformulate the sharp image reconstruction problem and incorporate the mask image M into Eq. (6). Besides, we also add in the sparse gradient constraint [12] to regularize the results when the weight for flash gradient constraint is low. The reconstruction problem is to minimize

$$\|I \otimes K - B\|^2 + \lambda_f M \circ \rho(\nabla I - \nabla F) + \lambda_i |\nabla I|^\alpha, \tag{10}$$

where $\circ$ denotes the pixel-wise multiplication operator. It can be solved using IRLS. As we can see from Fig. 3, by in-

troducing the mask M, artifacts in the reconstructed image can be successfully removed.

5. Experiments

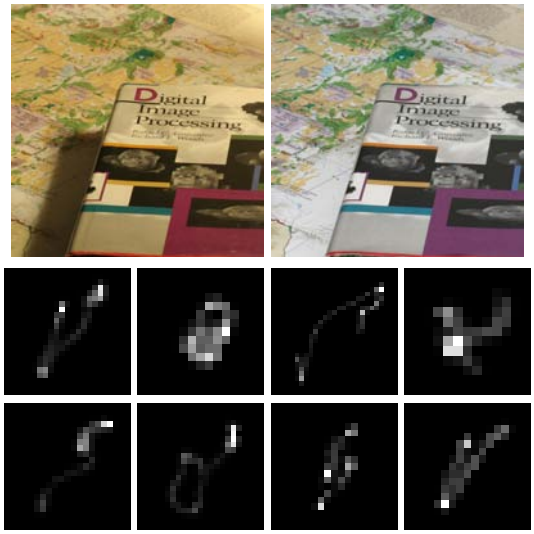
5.1. Image acquisition

A pair of blurred and flash images should be quickly taken one after another. We provide two ways to capture such an image pair. By connecting the camera to a computer, the first way is to use the computer to turn on/off the flash and take two images quickly. The second way is to take the two images using camera's high-speed capturing mode. Under high-speed capturing model, the time between two shots can be very small. In the first shot, the flash fires and we obtain the flash image, while in the second shot, the flash is not recharged in time. Thus, we get the blurred image. In practice, we use the second way. As the time between the two shots is small, the motion during the shots is basically a translation, which just causes a shift in the estimated blur kernel. Therefore, no image alignment is required.

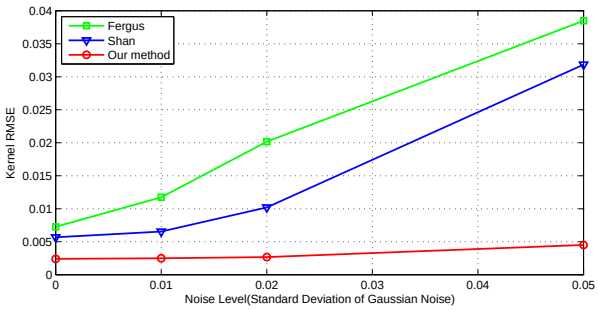
5.2. Results

We first test the accuracy of our kernel estimation using a set of synthetic images. Fig. 4(a) shows a sharp image I, its corresponding flash image F and a set of blur kernels created by Levin *et al*. [13]. I is blurred using the blur kernels to generate a series of blur images. After that, additive white Gaussian noises (AWGN) with different variances (0.01, 0.02 and 0.05) are added to the blurred images. We apply our kernel estimation method on these synthetic images, and quantitatively compared its performance with the kernel estimation methods proposed by Fergus *et al*. [6] and Shan *et al*. [17]. Both methods use only the blurred image to estimate the kernel. We fine tuned the parameters of both methods to generate good results. Fig. 4 (b) shows the root mean squared errors (RMSE) between estimated kernels and the ground truth kernels. When the noise level is low, all the three methods can obtain accurate kernels. The kernel error of our method is the smallest among them. However, kernel estimation using a single blurred image suffers from noise problem. As shown in the figure, when the noise level increases, the estimation error of the other two methods increases dramatically to a high level, while our method can still perform well and get accurate kernel estimation results.

In Fig. 5, we compared our method with three state-of-the-art single image deblurring algorithms: Fergus *et al*.'s method, Shan *et al*.'s method and fast motion deblurring. As the blur is large, their kernel estimation is not accurate enough, leading to some artifacts in the deblurred result. Some image details are lost in the process of reducing these artifacts. Generally, our method can obtain more accurate blur kernel and deblurring results with richer details and clearer structures. This is because the corresponding flash images provide robust constraints for both blur kernel estimation and sharp image reconstruction. Single image deblurring methods usually fail when the blur is large, but our method can handle very large motion blurs. One of the examples is shown in Fig. 8, the estimated blur kernel is 99×99. Our method robustly produces a accurate kernel and a high-quality sharp image.



(a) Sharp image I, flash image F, and blur kernels



(b) Kernel estimation error under different noise levels

Figure 4. Kernel estimation error. Existing kernel estimation methods using single image suffer from noise, while our method is robust to noise.

We then compare our method with Yuan *et al*.'s method [21]. As shown in Fig. 6, our method outperforms theirs in both kernel estimation and image reconstruction. Their kernel estimation is affected by noise and detail lost in the denoised image. In contrast, we use flash image, which provides a robust gradient constraint, which helps to estimate a accurate blur kernel and provides rich image details. Therefore, our result contains fine details. Similar results can be seen in Fig. 7, where we compare our method with dual motion deblurring [3]. These results show that an additional flash image is usually a better choice for image de-

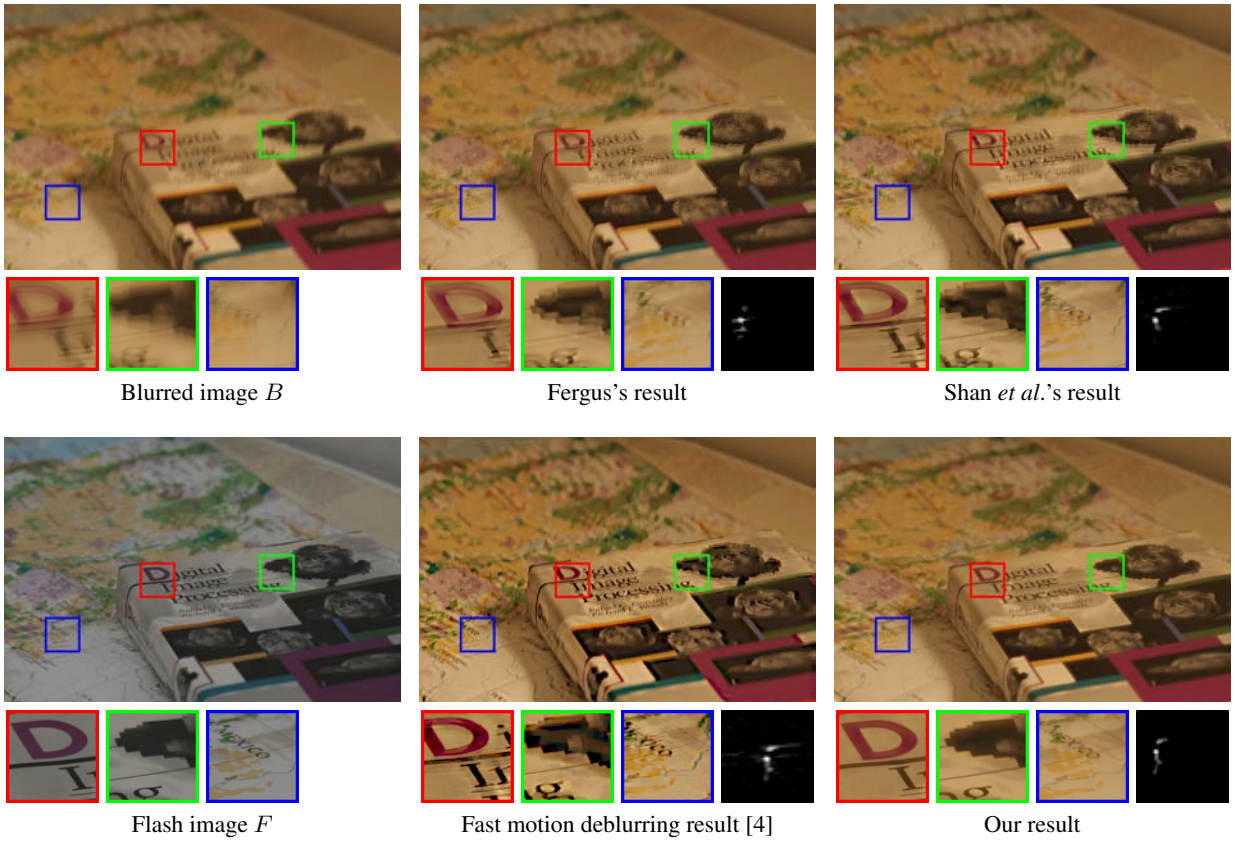


Figure 5. Comparison with single image deblurring methods. The performance of single image deblurring methods is affected by large blur, but our method is robust and can still obtain accurate kernel and reconstruction results with fine image details.

blurring compared with an additional noisy image or a second blurred image.

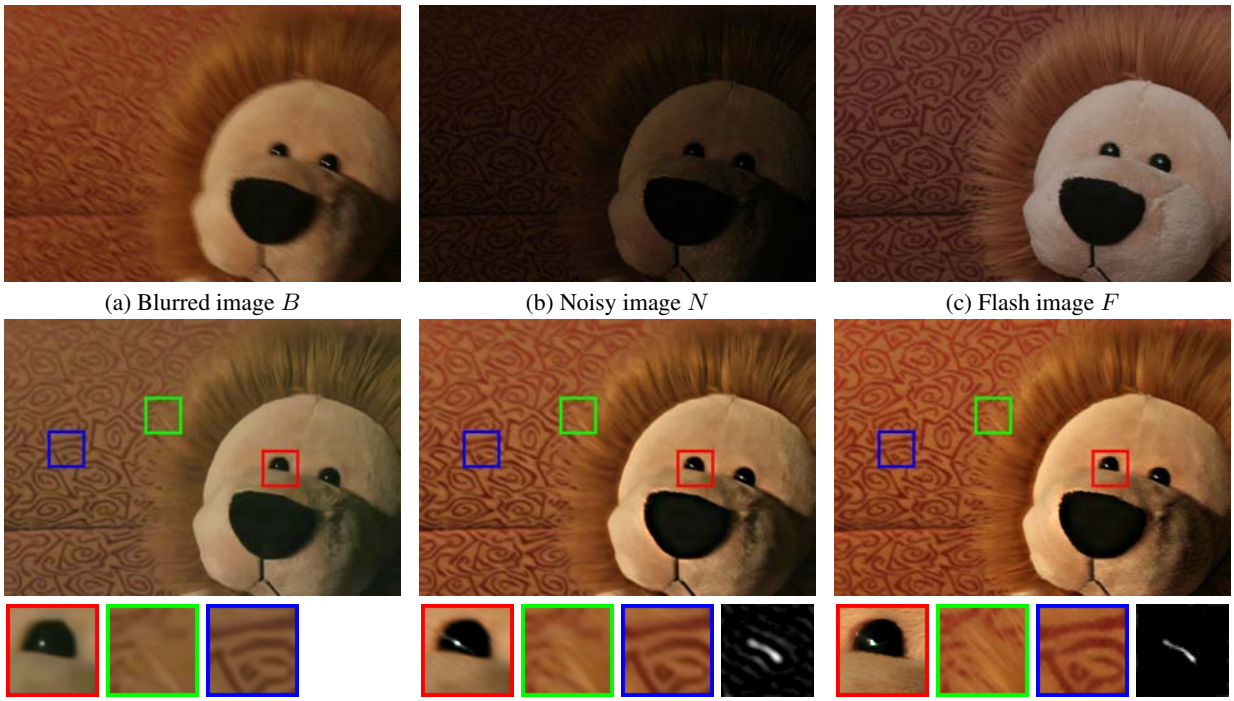
We also compare our method with color transfer method [19] in Fig. 9. The input blurred and flash images are shown in Fig. 1. The color transfer method only transfer the global light color from the blurred image to the flash image. However, the local lighting conditions can not be transferred. By comparison, our method can well preserve the lighting conditions presented in the blurred image.

6. Conclusion

We have proposed an image deblurring approach to remove camera motion blur using a pair of blurred and flash images. Our approach takes advantage of both images to produce a high quality reconstructed image. We propose the flash gradient constraint and integrate it into a MAP framework. We solve the optimization problem by iteratively performing blur kernel estimation and non-blind deconvolution. We demonstrate that our method outperforms existing single and multiple image deblurring methods.

By using a flash image, our method shares some common limitations of previous flash/no-flash techniques, *i.e.* it is more suitable for static indoor scenes. Besides, the spatially invariant motion blur model may not hold sometimes in practice. In the future, we plan to address this problem and extend our work to handle spatially variant motion blur or defocus blur.

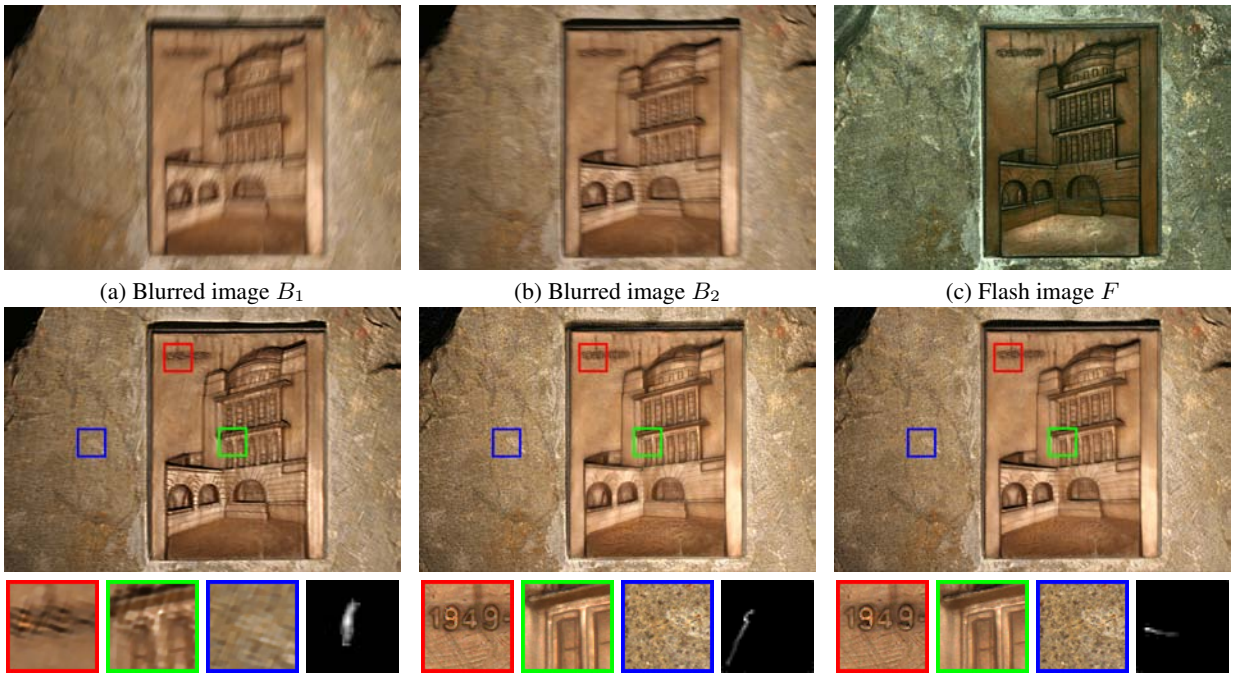
Figure 9. Comparison with color transfer. Our method is able to better preserve the lighting condition.



(a) Blurred image B (b) Noisy image N (c) Flash image F

(d) Denoised result of N (e) Yuan *et al.*'s result (f) Our result

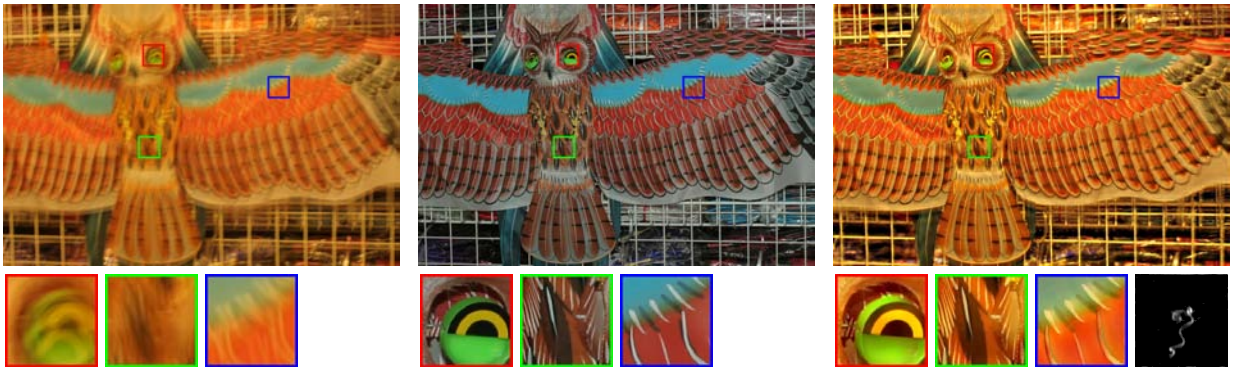
Figure 6. Comparison with blurred/noisy image deblurring. (d) shows the enhance noisy image. (e) Yuan *et al.*'s result using B and N. (f) Our result using B and F.



(a) Blurred image B_1 (b) Blurred image B_2 (c) Flash image F

(d) Dual motion deblurring result (e) Our result from B_1 and F (f) Our result from B_2 and F

Figure 7. Comparison with dual motion deblurring. (d) shows the dual motion deblurring result using B_1 and B_2. The blur kernel shown here is the estimated K_1. (e), (f) show our results using one of the blurred image and the flash image.



Blurred Image B Flash Image F Our Result

Figure 8. A real image example with very large blur. The size of the estimated blur kernel is 99×99.

Acknowledgment

We would like to thank Xiaopeng Zhang, Ning Ye, Hao Li for their insightful discussions, as well as Jia Chen for providing dual motion deblurring results. We acknowledge the generous support of NUS.

References

[1] A. Agrawal, R. Raskar, S. K. Nayar, and Y. Li. Removing photography artifacts using gradient projection and flash-exposure sampling. In *ACM Trans. Graphics*, pages 828–835, New York, NY, USA, 2005. ACM.

[2] M. Ben-Ezra and S. Nayar. Motion Deblurring using Hybrid Imaging. In *Proc. CVPR*, volume I, pages 657–664, 2003.

[3] J. Chen, L. Yuan, C.-K. Tang, and L. Quan. Robust dual motion deblurring. In *Proc. CVPR*, pages 1–8, 2008.

[4] S. Cho and S. Lee. Fast motion deblurring. In *SIGGRAPH Asia*, pages 1–8, 2009.

[5] E. Eisemann and F. Durand. Flash photography enhancement via intrinsic relighting. *ACM Trans. Graphics*, 23(3):673–678, 2004.

[6] R. Fergus, B. Singh, A. Hertzmann, S. T. Roweis, and W. T. Freeman. Removing camera shake from a single photograph. *ACM Trans. Graphics*, 25(3):787–794, 2006.

[7] P. Huber. *Robust Statistics*. Wiley, New York, 1974.

[8] J. Jia. Single image motion deblurring using transparency. In *Proc. CVPR*, pages 1–8, 2007.

[9] Q. Ke and T. Kanade. Robust l1 norm factorization in the presence of outliers and missing data by alternative convex programming. In *Proc. CVPR*, volume 1, pages 739–746, 2005.

[10] D. Krishnan and R. Fergus. Dark flash photography. In *ACM Trans. Graphics*, pages 1–11, 2009.

[11] D. Kundur and D. Hatzinakos. Blind image deconvolution. *IEEE Signal Processing Magazine*, 13(3):43–64, 1996.

[12] A. Levin, R. Fergus, F. Durand, and W. T. Freeman. Image and depth from a conventional camera with a coded aperture. In *ACM Trans. Graphics*, page 70, 2007.

[13] A. Levin, Y. Weiss, F. Durand, and W. T. Freeman. Understanding and evaluating blind deconvolution algorithms. In *Proc. CVPR*, pages 1964–1971, 2009.

[14] L. Lucy. An iterative technique for the rectification of observed distributions. *Astronomy*, 79:745–754, 1974.

[15] R. Neelamani, H. Choi, and R. Baraniuk. Forward: Fourier-wavelet regularized deconvolution for ill-conditioned systems. *IEEE Trans. Signal Process.*, 52(2):418–433, 2004.

[16] G. Petschnigg, R. Szeliski, M. Agrawala, M. Cohen, H. Hoppe, and K. Toyama. Digital photography with flash and no-flash image pairs. In *ACM Trans. Graphics*, pages 664–672, 2004.

[17] Q. Shan, J. Jia, and A. Agarwala. High-quality motion deblurring from a single image. *ACM Trans. Graphics*, 27(3):1–10, 2008.

[18] Y.-W. Tai, H. Du, M. S. Brown, and S. Lin. Image/video deblurring using a hybrid camera. In *Proc. CVPR*, pages 1–8, 2008.

[19] Y.-W. Tai, J. Jia, and C.-K. Tang. Local color transfer via probabilistic segmentation by expectation-maximization. In *Proc. CVPR*, volume 1, pages 747–754, 2005.

[20] Y. Wang, J. Yang, W. Yin, and Y. Zhang. A new alternating minimization algorithm for total variation image reconstruction. *SIAM J. on Imaging Sciences*, 1(3):248–272, 2008.

[21] L. Yuan, J. Sun, L. Quan, and H.-Y. Shum. Image deblurring with blurred/noisy image pairs. *ACM Trans. Graphics*, 26(3):1, 2007.

[22] L. Yuan, J. Sun, L. Quan, and H.-Y. Shum. Progressive inter-scale and intra-scale non-blind image deconvolution. *ACM Trans. Graphics*, 27(3):1–10, 2008.